

Day 44: Introduction to Operating System (OS) and Its Architecture

■ What is an Operating System (OS)?

An **Operating System (OS)** is **system software** that acts as an **interface** between users and hardware. It manages CPU, Memory, I/O devices, Files, Processes, and Security. Without an OS, a computer is just a collection of hardware with no coordination.

■ Functions of OS

■ Process Management	Creating, scheduling, and terminating processes
■ Memory Management	Allocating & deallocating RAM efficiently
■ File System Management	Organizing files on storage devices
■ I/O Device Management	Controlling and communicating with devices
■ Security & Protection	User authentication, permissions
■ User Interface	CLI (Command Line) or GUI (Graphical UI)

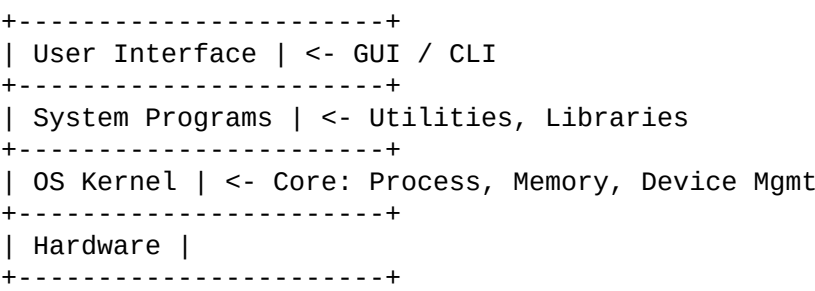
■■ Operating System Architecture

An OS is designed in layers to make it modular and easy to manage.

■ Types of OS Architecture

Monolithic Architecture	Entire OS runs in a single kernel space. Fast but harder to maintain/debug. Example: Linux.
Microkernel Architecture	Minimal kernel with only basic services in kernel mode, others in user space. More modular.
Layered Architecture	OS is divided into layers; each layer communicates with the next. Easy to debug.
Modular (Hybrid) Architecture	Combination of monolithic + microkernel. Loadable kernel modules. Example: Linux.

■■ OS Architecture Diagram



■ Key Takeaways

- OS is the bridge between user & hardware
- Manages resources: CPU, Memory, Devices, Files
- Different architectures exist: Monolithic, Microkernel, Layered, Hybrid
- Modern OS (Linux, Windows) mostly use modular/hybrid architecture for flexibility & performance